AWOL for Audio: Language-to-Sound Generation via Parametric Synthesis

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Mariagiusi Nicodemo

Abstract

This work explores linking textual descriptions to interpretable FM-synthesis controls, inspired by the AWOL paradigm. The pipeline progresses from a baseline MLP to a regularized version, and finally to a conditional RealNVP flow. Results show low parameter errors and improved semantic coherence with the stochastic flow, alongside smooth trajectories in interpolation and extrapolation.

1. Introduction

This project addresses the problem of linking textual descriptions to interpretable FM-synthesis controls, avoiding black-box approaches and preserving transparency in the parameter space. Inspired by the AWOL paradigm, we explore the direct mapping of CLAP text embeddings into the control space of an FM synthesizer.

The approach progresses from baseline and regularized MLPs to a conditional RealNVP flow, capable of capturing richer distributions and producing smoother trajectories. The analysis also extends to latent-space exploration through interpolation and extrapolation between text embeddings, revealing fluid transitions and coherence even out of distribution.

Overall, the results confirm the feasibility of an interpretable language-to-sound mapping, while highlighting both limitations and promising directions for future work.

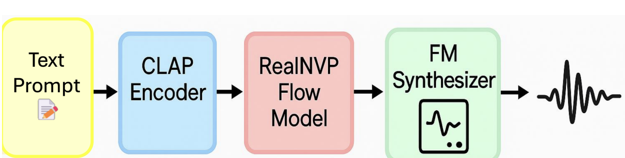

Figure 1. From text to CLAP embeddings, through the predictive model, to normalized FM-synthesis controls.

Email: Mariagiusi Nicodemo <nicodemo.2114171@studenti.uniroma1.it>.

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All materials (code, notebooks, checkpoints, and sample audio) are available at: `https://github.com/Mariagiusi23/ID-001-AWOL-for-Audio`.

2. Related Work

This work draws inspiration from the **AWOL** paradigm (Zuffi & Black, 2024), which maps semantic embeddings into interpretable parameters without differentiable renderers, here extended from 3D to audio. For text–audio representation we adopt **CLAP** (Elizalde et al., 2022; Wu et al., 2022), used both as conditioning and as an evaluation metric; related methods such as **AudioCLIP** (Guzhov et al., 2021) and **CLIP** (Radford et al., 2021) confirm its effectiveness.

On the modeling side, **normalizing flows** balance flexibility and tractability, with **RealNVP** (Dinh et al., 2017) as the main choice, alongside **VAEs** (Kingma & Welling, 2014) and **Glow** (Kingma & Dhariwal, 2018). Finally, end-to-end systems like **DDSP** (Engel et al., 2020) and **MusicGen** (Copet et al., 2023) achieve strong perceptual quality but sacrifice interpretability, which remains central in this work.

3. Method

The goal is to learn a function that maps a CLAP text embedding $z = \mathrm{CLAP}(t)$ into a vector of normalized FM-synthesis controls y:

$$f : z \in \mathbb{R}^d \longmapsto y \in [0,1]^k.$$

Baselines. As a starting point, a fully-connected MLP was implemented to learn the direct CLAP→controls mapping. Training minimizes the mean squared error (MSE) between targets and predictions:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N}\sum_{i=1}^{N} \|y_i - \hat{y}_i\|^2.$$

This baseline confirmed the feasibility of the mapping, though with limited stability. A regularized version with dropout and L2 penalty was then introduced, improving generalization and producing more consistent predictions.

Contribution. The next step employed a conditional normalizing flow (RealNVP), which captures multimodal distributions and generates smoother control trajectories, overcoming the limitations of purely deterministic mappings.
Finally, the method was extended to latent-space exploration through interpolation and extrapolation between text embeddings. For two prompts z_A and z_B, we consider linear combinations:

$$z_\alpha = (1-\alpha)z_A + \alpha z_B, \quad \alpha \in \mathbb{R}.$$

These experiments revealed fluid sound transitions and the model's ability to preserve coherence even out of distribution, highlighting the added value of the proposed contribution compared to the baselines.

4. Experiments and Results

Evaluation combined quantitative metrics with qualitative analyses. Root mean squared error (RMSE) was used to measure numerical accuracy on synthesis controls, while CLAP similarity provided a perceptual proxy for semantic coherence between prompts and generated audio.
Table 1 reports RMSE values for individual parameters. Amplitude, carrier frequency, and modulation index are predicted with errors below 0.12, indicating a stable mapping. The frequency ratio is harder to estimate (RMSE $\approx$ 0.18), reflecting its critical role in FM synthesis where small variations strongly affect timbre.

Parameter	RMSE
Amplitude	0.068
Carrier Frequency	0.114
Ratio	0.180
Modulation Index	0.098

Table 1. RMSE per parameter on the test set.

Table 2 shows CLAP similarity across modes. The MLP and zero modes remain stable around 0.08–0.09 regardless of τ, consistent with deterministic behavior. The stochastic flow peaks at $\tau \approx 0.6$ (0.164) but degrades at $\tau = 1.5$ (0.081).

Mode	τ	CLAP sim
mlp	0.2–1.0	$\approx$ 0.089
zero	0.2–1.0	$\approx$ 0.080
sample	0.6	0.164
sample	1.5	0.081

Table 2. CLAP similarity across models and temperatures.

Further evaluation considered unseen prompts and latent-space exploration. On unseen inputs, the model produced coherent and plausible controls without collapse, indicating that it learned a generalizable mapping rather than memorizing training data. Interpolations between text embeddings yielded smooth transitions in both controls and generated sounds, while extrapolation ($\alpha < 0$ or $\alpha > 1$) largely preserved coherence.
Overall, results confirm the validity of the proposed method: baselines provide stability, the conditional flow improves semantic alignment, and latent exploration reveals the regularity of the mapping. Limitations remain, such as low absolute CLAP similarity values and difficulty with the frequency ratio, but the overall picture is positive and encouraging.

5. Discussion and Conclusions

This work confirmed the feasibility of directly mapping CLAP text embeddings to interpretable FM-synthesis controls. The baselines demonstrated stability, while the conditional flow improved semantic coherence and yielded smoother trajectories. Latent-space interpolations and extrapolations further highlighted the continuity and regularity of the mapping.
Clear limitations remain: absolute CLAP similarity values are relatively low, and the frequency ratio is harder to predict with precision. In addition, some extra experiments—attempted as extensions of the fourth notebook—revealed technical instabilities that could not be fully investigated within this work but represent meaningful directions for future research.
Overall, the project provides a solid and interpretable demonstration of connecting language to sound through structured parameters, paving the way for more refined approaches to controlled audio generation.

Appendix

Additional experiments were carried out but not included in the main results. A first set involved hyperparameter search: dropout, L2 regularization, and learning rate were varied, showing that excessive regularization stabilized the model but reduced adaptability.
Further trials with the conditional flow, especially an extension of the fourth notebook, led to numerical instabilities and incoherent outputs. While inconclusive, these attempts highlighted current limitations and the need for deeper study.
Finally, some latent-space extrapolations ($\alpha \ll 0$ or $\alpha \gg 1$) produced uninterpretable controls, indicating that the model is most reliable in local regimes. These failed attempts also clarified the boundaries of the method and point to future research directions.

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